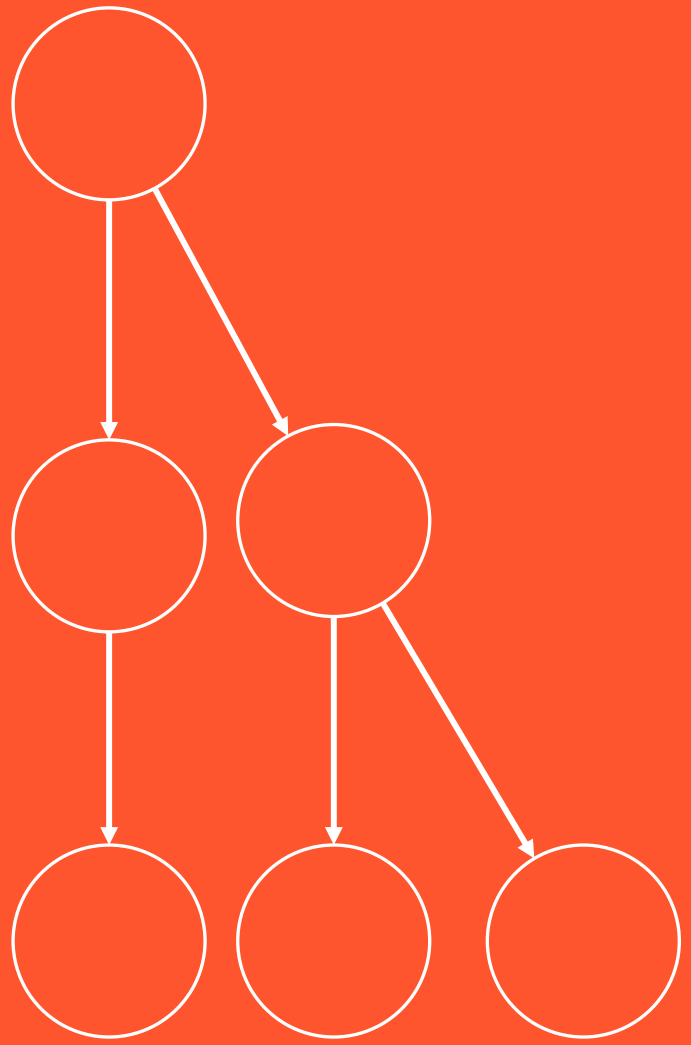


BCOG 100

Neural Circuits, Valence, and Affect

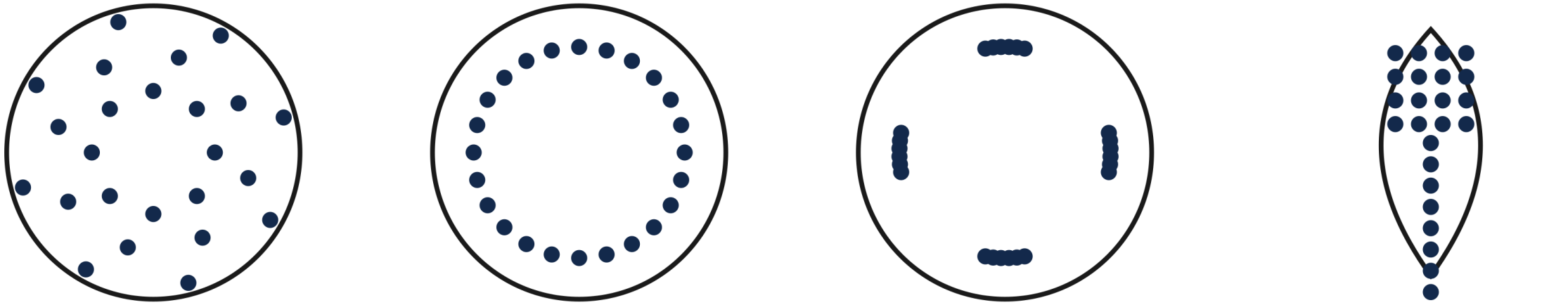
Lecture 4.3 — The Biology of Neural Circuits

UNIVERSITY OF ILLINOIS URBANA-CHAMPAIGN

Why the Brain Is in the Head

A net thickens into a **nerve ring**, then into a **ganglion** — a center without being a brain. In bilaterians the largest ganglia end up at the front, with the sensors: **cephalization**. Nothing is *added* along the row.

The same twenty-four neurons, gathered four ways

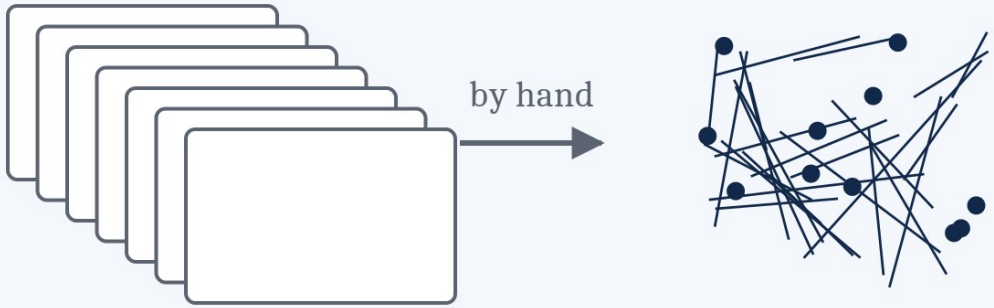


Nothing is added along the row. Count them.

One Animal We Know Completely

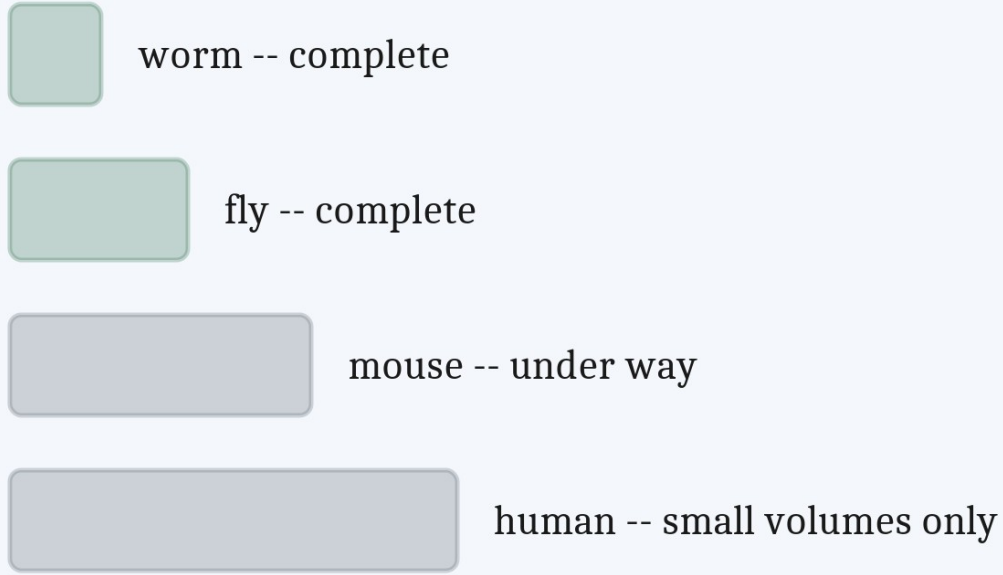
302 neurons in the adult hermaphrodite, and every connection between them reconstructed by hand from electron micrographs of serial slices. A **connectome** is that: every cell, and every connection.

how the map was made



302	neurons
several thousand	connections
over a decade	to produce

and how far the enterprise has got

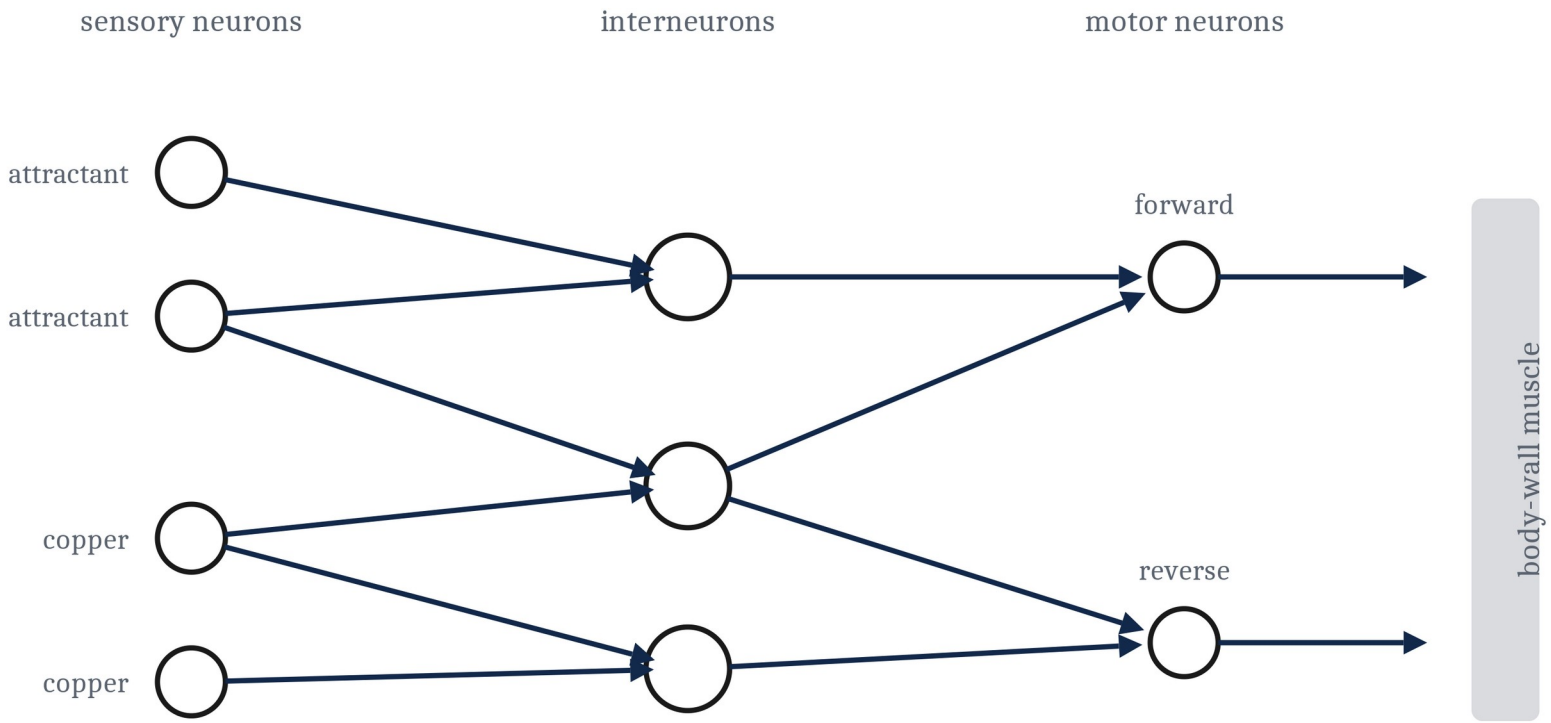


a wiring diagram cannot be read off an animal. it has to be built.

Sense, Combine, Act — in Three or Four Cells

Sensory neurons in the head, to interneurons receiving from several at once, to motor neurons driving the body wall. The worm does not point itself at the food. **Going straight is not a behavior — it is what happens when the rule for turning is not met.**

One real circuit that steers

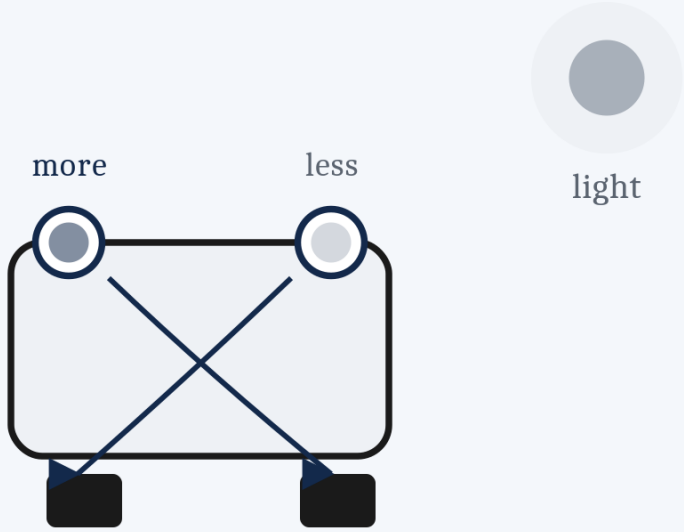


Which route a signal feeds is what makes it good news or bad.

Two Ways to Steer

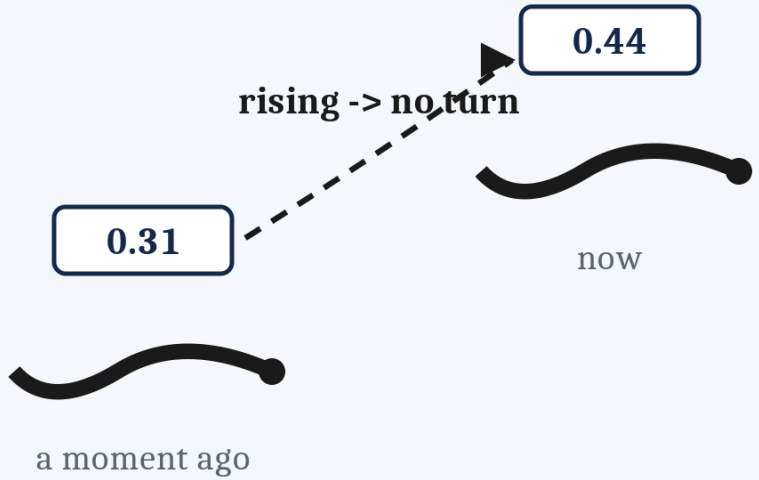
The vehicle you built compares **two places at one instant** — its body's geometry delivers the answer. Each side of the worm compares **two moments at one place**: now against a moment ago. The vehicle is told which way the light is. The worm is not.

the vehicle: two places, one instant



the body's geometry gives the answer

the worm: two moments, one place



nothing tells it which way.

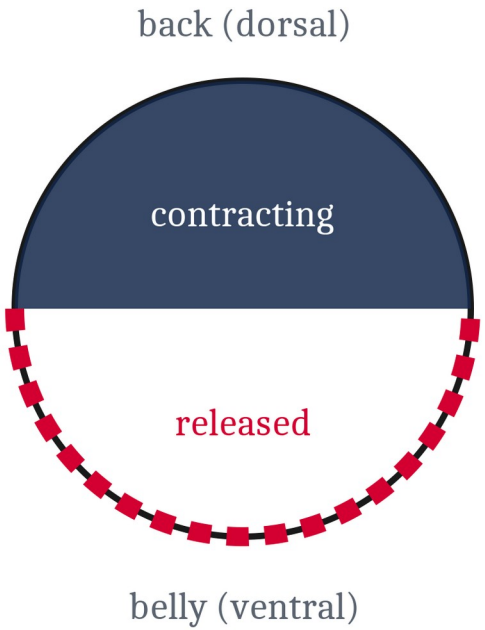
a millimeter of body buys no left-right comparison

In both, which route a signal feeds is what makes that signal good news or bad.

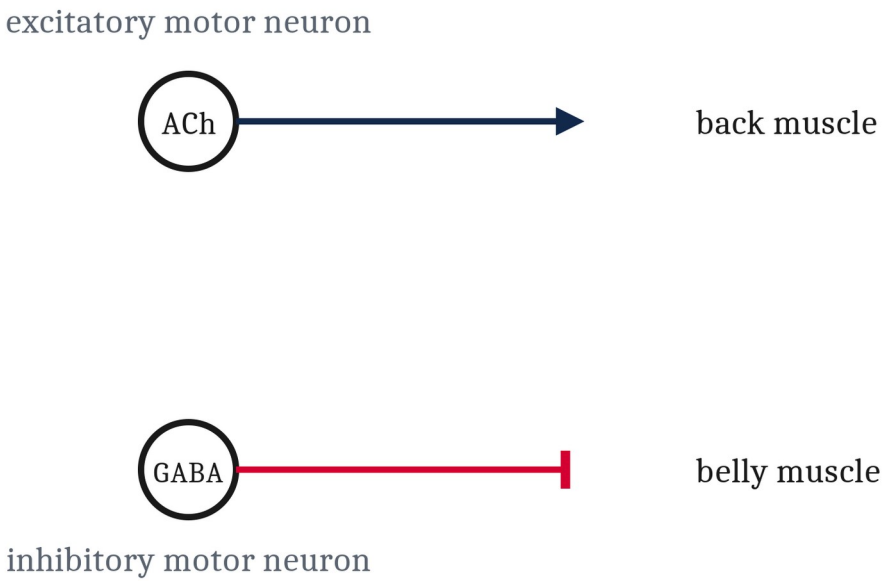
The Two Ends We Kept Skipping

Transduction — converting an event in the world into a change in membrane voltage — is the one thing a sensory neuron does that no other neuron does. At the far end, the **neuromuscular junction**: the verdict stops being a voltage and becomes a force.

The body wall runs on mutual inhibition



one cross-section

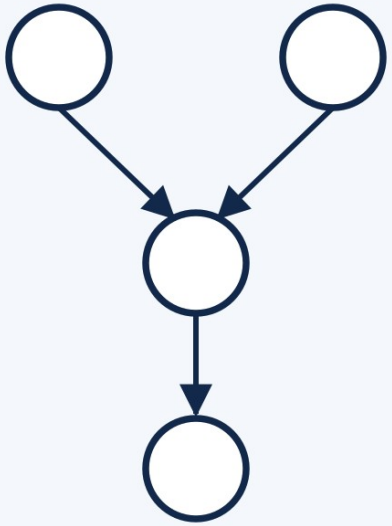


Then the two swap, and the alternation runs head to tail.

The Map Is Complete, and It Is Not Enough

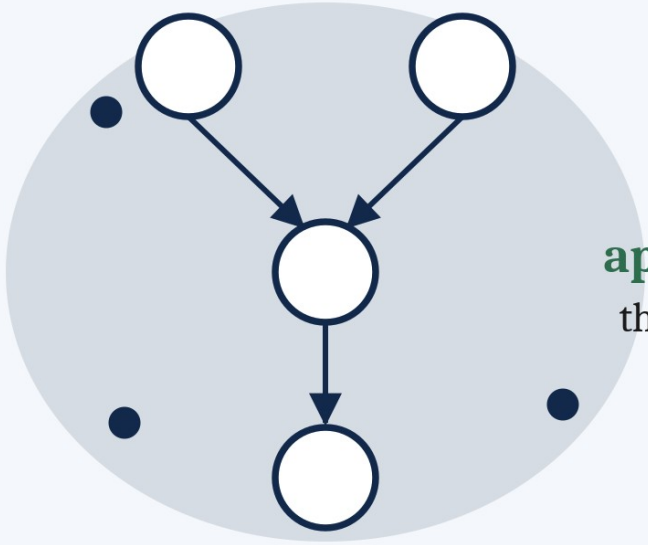
The worm's wiring has been known for decades and its behavior still cannot be predicted from the diagram. A fed worm and a hungry worm respond to the same molecule in opposite ways — **same 302 neurons, wired the same way**.

fed



avoids
the CO2

hungry



approaches
the same CO2

a chemical released into the tissue,
not into a synapse

Same 302 neurons, wired the same way. The connections are not what changed.

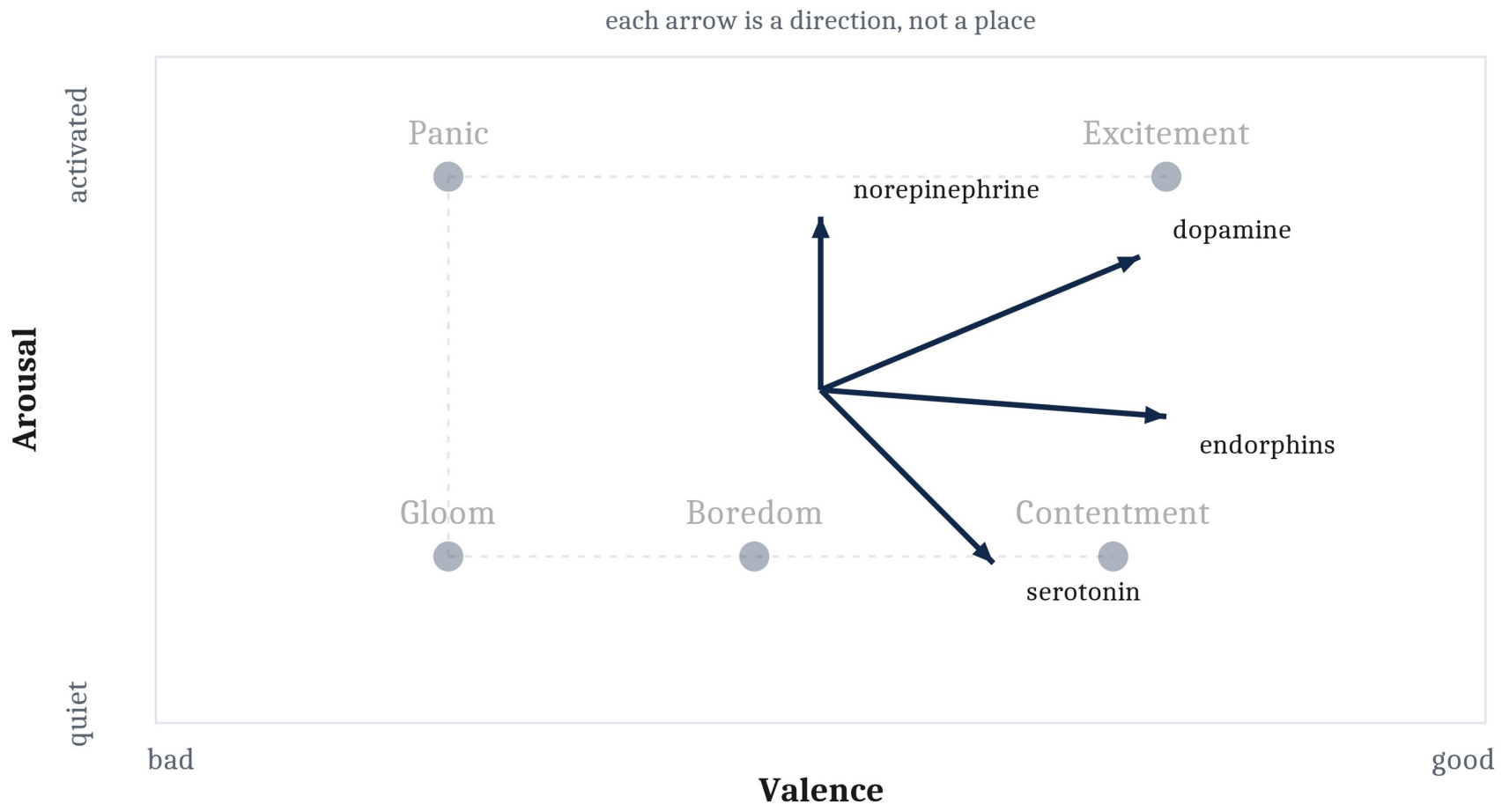
Four Molecules

Not four unrelated chemicals — four ways of moving an animal around the same two-dimensional space we built in the last lecture. **Read the third column as a direction of travel, not a destination.**

Molecule	What it does	Where it moves the animal	In human life
Dopamine	Drives pursuit of a thing: wanting, and the effort spent getting it	Raises arousal; raises *expected* valence rather than current valence	Motivation, and what is captured in addiction
Serotonin	Signals satiety and sets the overall tone of the system	Lowers arousal; raises valence toward contentment	Mood and patience; a common target of medication
Norepinephrine	Sets vigilance: how strongly to react to anything	Raises arousal, at either valence	Alertness and startle; arousal set too high in anxiety
Endorphins and other opioids	Carry relief and satisfaction: liking rather than wanting	Raises valence; lowers arousal	Pleasure and pain relief; the target of opioid drugs

Directions, Not Locations

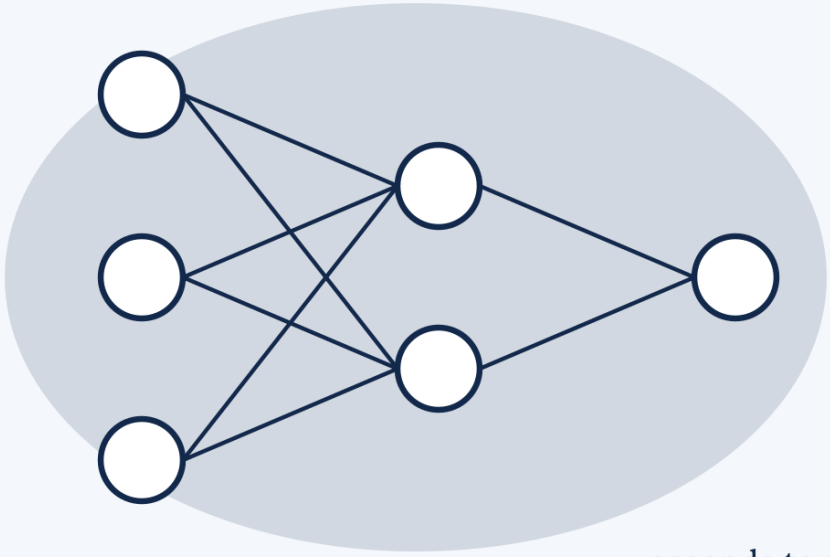
The same affect plane, with the four drawn as **arrows from a common point** rather than as positions. A neuromodulator does not put an animal somewhere — it moves the animal from wherever it already is.



Where the Comparison Finally Breaks

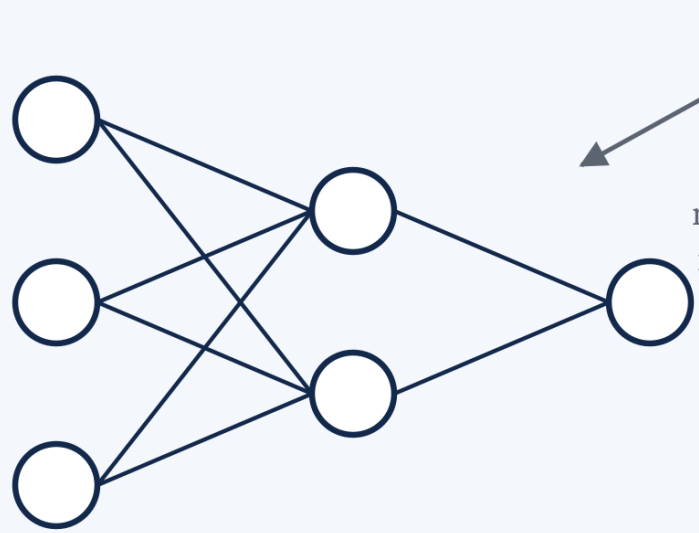
An artificial network has weights and activations, and that is all it has. Nothing corresponds to a chemical released into the space between units, reaching all of them at once, and then **wearing off**.

a biological circuit



seconds to minutes,
and then it fades

an artificial network



set from outside,
not released by the
network into itself

no chemical between units

no mechanism for reaching all at once

nothing that wears off

The asymmetry is real. Whether it is a deficiency is not something anyone can currently say.

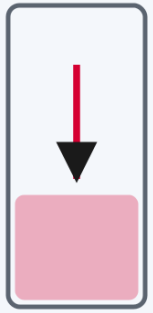
What a State Costs

A chemical in the tissue keeps a circuit tuned one way for minutes — which has a consequence the wiring diagram does not show. Staying keyed up is **spending**; stopping responding is **conserving**.

staying keyed up

still fleeing after the scent has thinned

still searching after the food is gone

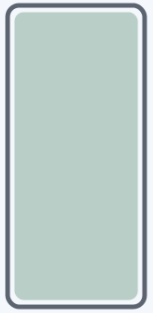


SPENDING

reserve

stopping responding

under conditions it cannot escape



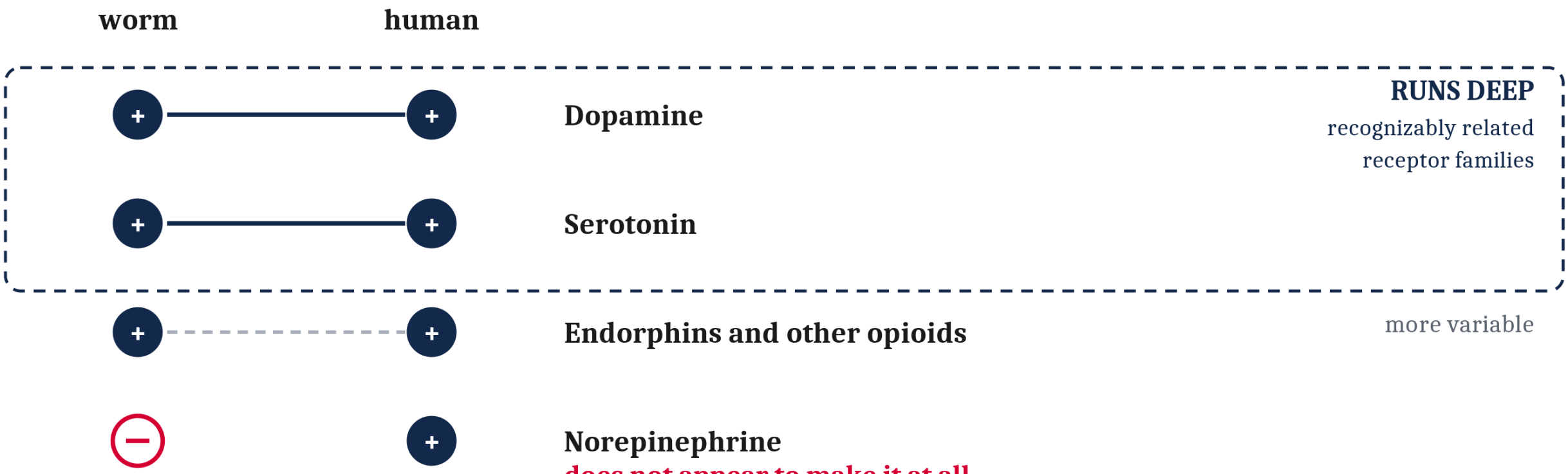
CONSERVING

reserve

Giving up is a thing bodies do to save energy, not simply a thing that goes wrong.

How Much of This Is Actually Ours

Dopamine and serotonin are present in both, acting through recognizably related receptor families. **The worm does not appear to make norepinephrine at all.** Two run deep; the others are more recent or more variable.

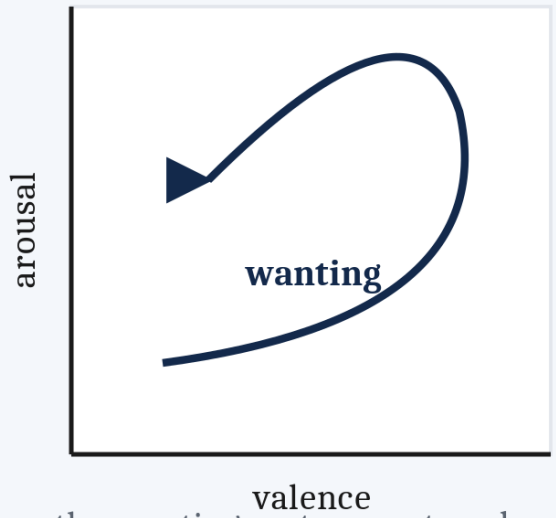


Not four ancient constants. Two run deep; the others are more recent or more variable.

Motivation, Mood, and Alertness

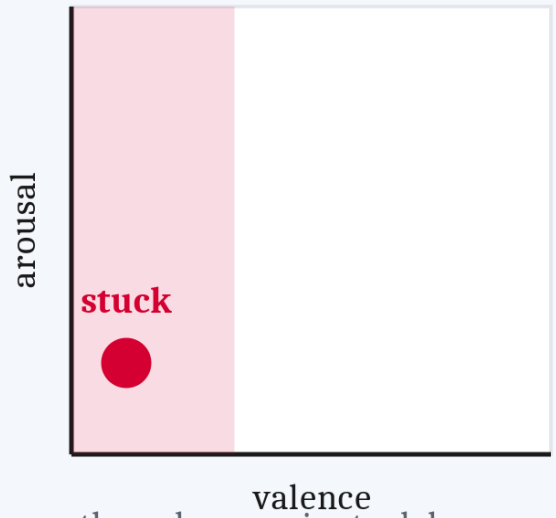
Addiction as the **wanting system captured**. Anhedonia as the **valence axis stuck low**. Anxiety as **arousal set high and staying there**. None of these is a complete account — and none of them is a simple chemical shortage.

addiction



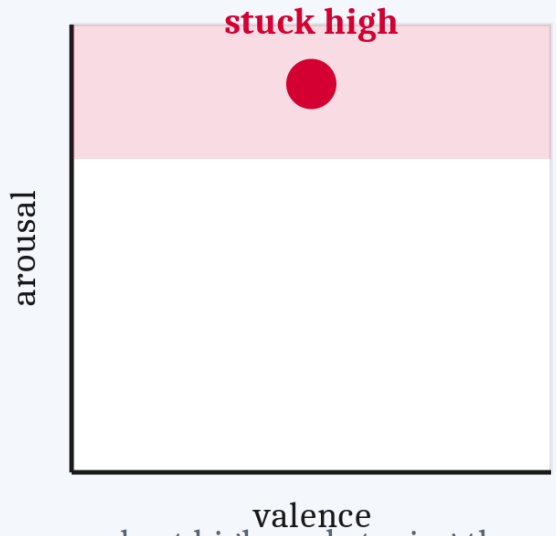
the wanting system captured

anhedonia



the valence axis stuck low

anxiety



arousal set high, and staying there

Understandable as. Not a definition, and not a diagnosis.

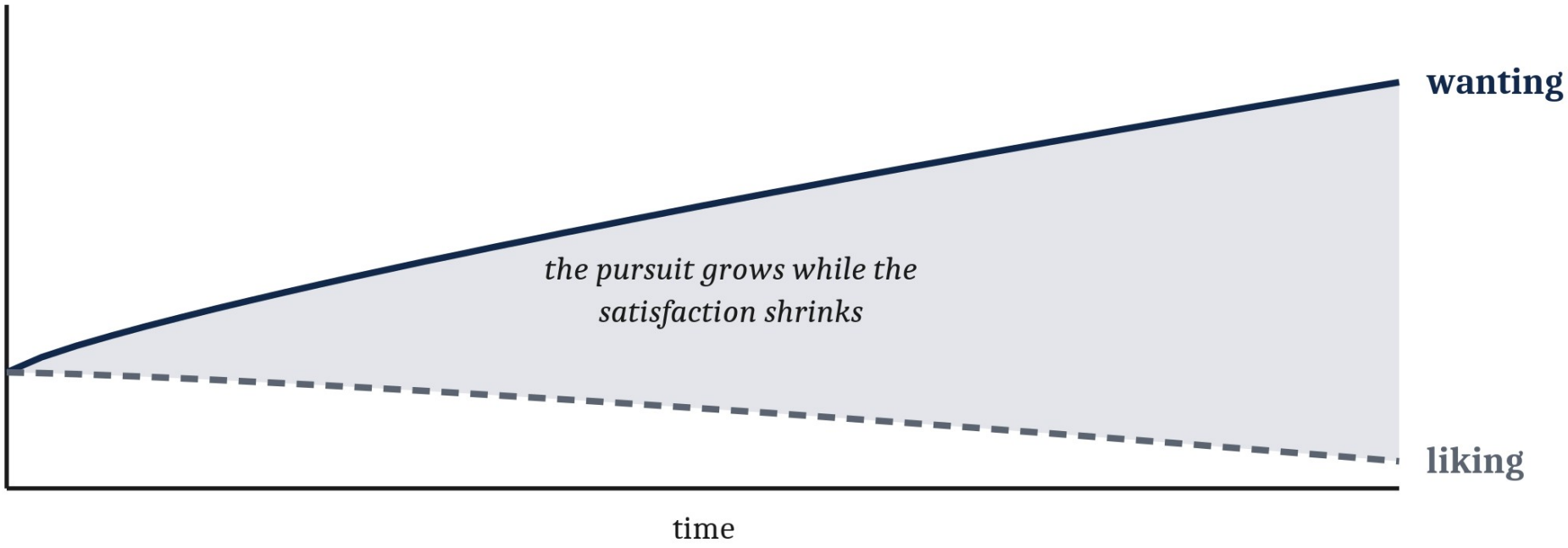
None is a complete account. None is a simple chemical shortage.

A System That Assigns Value Can Be Fooled About Value

Anything acting directly on the wanting chemistry produces **the pull toward a thing, without the thing**. When wanting and liking come apart, the pursuit can grow while the satisfaction shrinks.



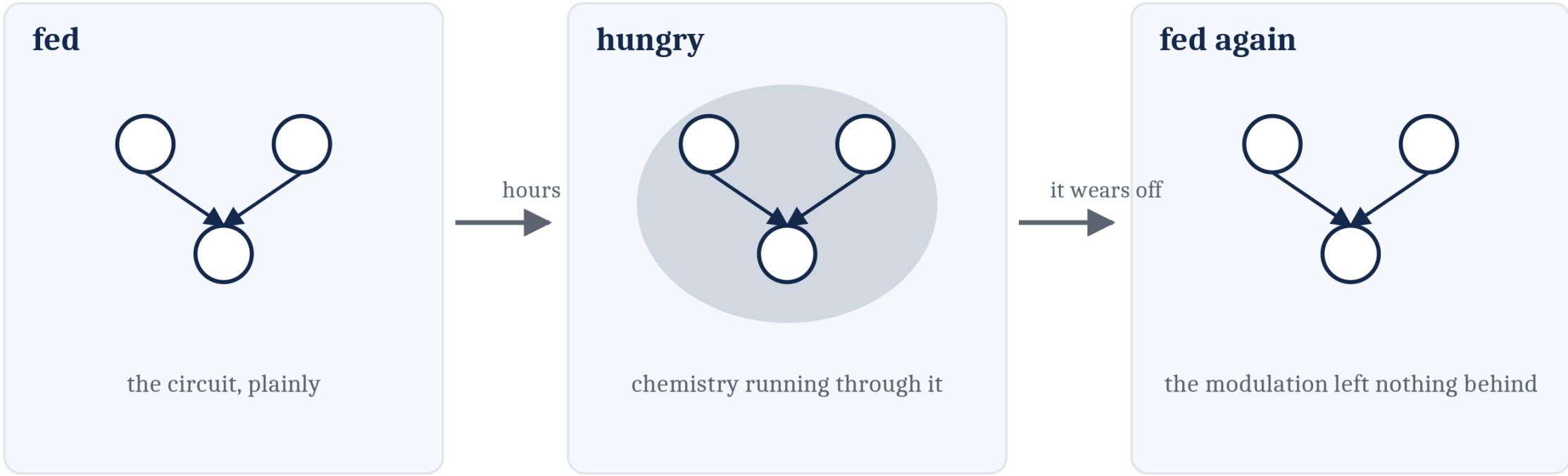
the pull toward a thing,
without the thing



And it is not only substances.

What Wiring Made Possible, and What It Still Cannot Do

Good and bad are **verdicts an arrangement assigns**. The neuromodulation that looked like the missing flexibility is not learning — it leaves nothing behind. But **the fixed weights are ours, not the animals'.** Real animals learn, these ones included.



the weights have not moved. nothing about that episode is kept.

how a weight changes within **a single lifetime: Module 5**

Real animals DO learn, these ones included. Holding the weights still is our simplification.

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